Properties of equality and congruence

ANSWER KEY

Core Concepts: Properties of equality and congruence

1 In $\triangle ABC$ studying properties of equality and congruence, $m\angle A = 39^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 96° ✓

b. 106°

c. 88°

d. 141°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (39^\circ + 45^\circ) = 96^\circ$.

2 A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length?

a. 14 cm

b. 16 cm ✓

c. 24 cm

d. 32 cm

Multiplying the original length by the scale factor gives $8 \times 2 = 16$ cm.

3 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 1$?

a. 5

b. 10

c. 8 ✓

d. 11

Substituting $x = 1$ into $y = 3x + 5$ gives $y = 3(1) + 5 = 8$.

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Procedural Fluency & Algebraic Manipulation

4 In $\triangle ABC$ studying properties of equality and congruence, $m\angle A = 39^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 89° ✓

b. 99°

c. 81°

d. 141°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (39^\circ + 52^\circ) = 89^\circ$.

5 A geometric figure scaled by factor $k = 3$ has an original side length of 8 cm. What is the corresponding image length?

a. 21 cm

b. 24 cm ✓

c. 32 cm

d. 72 cm

Multiplying the original length by the scale factor gives $8 \times 3 = 24$ cm.

6 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 2$?

a. 8

b. 13

c. 11 ✓

d. 14

Substituting $x = 2$ into $y = 3x + 5$ gives $y = 3(2) + 5 = 11$.

Graphical & Geometric Interpretation

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7 In $\triangle ABC$ studying properties of equality and congruence, $m\angle A = 39^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 82° ✓

b. 92°

c. 74°

d. 141°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (39^\circ + 59^\circ) = 82^\circ$.

8 A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length?

a. 28 cm

b. 32 cm ✓

c. 40 cm

d. 128 cm

Multiplying the original length by the scale factor gives $8 \times 4 = 32$ cm.

9 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 3$?

a. 11

b. 16

c. 14 ✓

d. 17

Substituting $x = 3$ into $y = 3x + 5$ gives $y = 3(3) + 5 = 14$.

Exam-Style Problem Solving & Modeling

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10 In $\triangle ABC$ studying properties of equality and congruence, $m\angle A = 39^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 75° ✓

b. 85°

c. 67°

d. 141°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (39^\circ + 66^\circ) = 75^\circ$.

11 A geometric figure scaled by factor $k = 5$ has an original side length of 8 cm. What is the corresponding image length?

a. 35 cm

b. 40 cm ✓

c. 48 cm

d. 200 cm

Multiplying the original length by the scale factor gives $8 \times 5 = 40$ cm.

12 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 4$?

a. 14

b. 19

c. 17 ✓

d. 20

Substituting $x = 4$ into $y = 3x + 5$ gives $y = 3(4) + 5 = 17$.

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